Ali Tleis

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Security Clearance: DoD Investigation in Progress

Education

Northeastern University — Boston, MA
Bachelor of Science in Computer Science and Sociology — *Expected Aug. 2028*
- **Coursework:** Algorithms & Data Structures, Object-Oriented Design, Artificial Intelligence, Discrete Structures

Experience

MIT Lincoln Laboratory — Lexington, MA
Web Application Developer (AI Integration) Co-op — *Jun. 2026 – Present*
- Shipped internal AI search used by **4,500+ engineers**, owning the technical proposal, implementation, and production release of Java/Spring Boot APIs and Python/FastAPI services.
- Cut query latency **70%** and model token consumption **40%** by combining hybrid Solr keyword/vector retrieval with Sentence Transformers reranking in Python/FastAPI.
- Preserved document access controls in search results and generated answers with permission-aware retrieval tied to existing identity services and incremental Norconex ingestion.
- Deployed Docker containers to existing Kubernetes infrastructure with Helm and GitLab CI/CD, adding automated integration tests and Prometheus/Grafana dashboards to support rollout and diagnose failures.
- Integrated internally hosted Llama inference through vLLM with streamed responses, retrieval-grounded prompts, and Redis caching for repeated queries.

Top Choice Realty — Staten Island, NY
Frontend Developer Intern — *Jun. 2025 – Sep. 2025*
- Reduced manual reconciliation time **30%** by automating MongoDB synchronization with Python and C# Azure Functions and refactoring schemas to prevent duplicate client and agent records.
- Implemented role-based access and document uploads in Express APIs backed by MongoDB Atlas, storing uploaded files in Azure Blob Storage.
- Added GitHub Actions workflows for automated builds, API tests, and deployment checks across the application.
- Delivered Next.js client-intake and agent-dashboard workflows using reusable React/TypeScript components, standardizing UI behavior and incorporating agent feedback into production releases.

Robert DeFalco Realty — Staten Island, NY
Computer Technician Intern — *Jun. 2023 – Sep. 2023*
- Automated provisioning for **20+ systems** using PowerShell, standardizing software installation, account setup, and Windows/Linux workstation configuration.
- Built a Python inventory utility to flag missing software, outdated systems, and configuration differences while supporting Microsoft Intune enrollment and Active Directory account administration.

Projects

EternalMonitor | eternalmonitor.dev | GitHub — Apr. 2026 – Present
- Owned AMD GPU support end to end for an app with **500+ users** and **175+ GitHub stars**, implementing Rust/FFmpeg AMF H.264/HEVC encoding, keyframe recovery, and software fallback.
- Validated AMD streaming on physical Windows and iPad hardware across DXGI capture, GPU encoding, Wi-Fi/USB transport, VideoToolbox decoding, and Metal rendering.

Upflow2x | upflow2x.com | GitHub — Jan. 2026 – Present
- Built a Python and Lua plugin for DaVinci Resolve that finds held frames in video, draws the missing in-betweens with RIFE, and upscales **2x** with Real-ESRGAN on the GPU via ONNX Runtime and DirectML.
- Wrote **750+** headless tests against a simulated Resolve API, run in CI on Linux, macOS, and Windows.

Technical Skills

Languages: Java, Python, Rust, C++, C#, TypeScript, JavaScript, SQL, PowerShell, Lua
Frameworks: Spring Boot, FastAPI, Node.js/Express, React, Next.js
Databases & Search: Apache Solr, Redis, MongoDB Atlas, PostgreSQL
AI: Sentence Transformers, vLLM, Llama, Gemini API
Infrastructure: Azure Functions, Azure Blob Storage, Docker, Kubernetes, Helm, Linux
Tools: GitLab CI/CD, GitHub Actions, Prometheus, Grafana